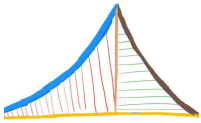
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE 2026 MATHEMATICS

A. Formulae

1.1.1 For a non-singular $n \times n$ matrix $\mathbf{A}$,

$$\mathbf{A}\,\operatorname{adj}\mathbf{A} = \left|\mathbf{A}\right|\mathbf{I}, \quad \mathbf{A}^{-1} = \frac{\operatorname{adj}\mathbf{A}}{\left|\mathbf{A}\right|} \tag{1.1.1.1}$$

$$\left|\operatorname{adj}\mathbf{A}\right| = \left|\mathbf{A}\right|^{n-1} \tag{1.1.1.2}$$

Solution: Taking the determinant on both sides of the first equation in (1.1.1.1),

$$\left|\mathbf{A}\right|\left|\operatorname{adj}\mathbf{A}\right| = \left|\left|\mathbf{A}\right|\mathbf{I}\right| = \left|\mathbf{A}\right|^{n} \tag{1.1.1.3}$$

which yields (1.1.1.2) when $\left|\mathbf{A}\right| \neq 0$.

1.1.2 The linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ with $\left|\mathbf{A}\right| \neq 0$ has the unique solution

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} \tag{1.1.2.1}$$

1.1.3 The area of the triangle with vertices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ is

$$\frac{1}{2}\left\|(\mathbf{B}-\mathbf{A}) \times (\mathbf{C}-\mathbf{A})\right\| \tag{1.1.3.1}$$

In $\mathbb{R}^2$ this reduces to $\frac{1}{2}\left|\left|\mathbf{B}-\mathbf{A} \quad \mathbf{C}-\mathbf{A}\right|\right|$ and the points are collinear if and only if

$$\operatorname{rank}\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix} = 1 \iff \left|\mathbf{B}-\mathbf{A} \quad \mathbf{C}-\mathbf{A}\right| = 0 \tag{1.1.3.2}$$

1.1.4 For any vectors $\mathbf{a}, \mathbf{b}$,

$$\left\|\mathbf{a}-\mathbf{b}\right\|^2 = \left\|\mathbf{a}\right\|^2 + \left\|\mathbf{b}\right\|^2 - 2\mathbf{a}^\top\mathbf{b} \tag{1.1.4.1}$$

$$\mathbf{a}^\top\mathbf{b} = 0 \iff \mathbf{a} \perp \mathbf{b} \tag{1.1.4.2}$$

1.1.5 The angle θ between the lines $\mathbf{x} = \mathbf{A}_1 + \lambda\mathbf{m}_1$ and $\mathbf{x} = \mathbf{A}_2 + \mu\mathbf{m}_2$ is given by

$$\cos\theta = \frac{\mathbf{m}_1^\top\mathbf{m}_2}{\left\|\mathbf{m}_1\right\|\left\|\mathbf{m}_2\right\|} \tag{1.1.5.1}$$

1.1.6 The foot of the perpendicular from $\mathbf{Q}$ to the line $\mathbf{x} = \mathbf{P}_0 + \lambda\mathbf{m}$ is $\mathbf{F} = \mathbf{P}_0 + \lambda_0\mathbf{m}$, where

$$\lambda_0 = \frac{\mathbf{m}^\top(\mathbf{Q}-\mathbf{P}_0)}{\mathbf{m}^\top\mathbf{m}} \tag{1.1.6.1}$$

Solution: $\mathbf{F}-\mathbf{Q}$ is orthogonal to $\mathbf{m}$. From (1.1.4.2),

$$\mathbf{m}^\top(\mathbf{P}_0 + \lambda_0\mathbf{m} - \mathbf{Q}) = 0 \implies \lambda_0 = \frac{\mathbf{m}^\top(\mathbf{Q}-\mathbf{P}_0)}{\mathbf{m}^\top\mathbf{m}} \tag{1.1.6.2}$$

1.1.7 The shortest distance between the skew lines $\mathbf{x} = \mathbf{A}_1 + \lambda\mathbf{m}_1$ and $\mathbf{x} = \mathbf{A}_2 + \mu\mathbf{m}_2$ is

$$d = \frac{\left|(\mathbf{A}_2-\mathbf{A}_1)^\top(\mathbf{m}_1 \times \mathbf{m}_2)\right|}{\left\|\mathbf{m}_1 \times \mathbf{m}_2\right\|} \tag{1.1.7.1}$$

1.1.8 For events A, B and a partition A_1, A_2, A_3 of the sample space with $\Pr(A_i) = p_i$, $\Pr(E|A_i) = q_i$,

$$\Pr(A|B) = \frac{\Pr(AB)}{\Pr(B)}, \quad \Pr(A+B) = \Pr(A) + \Pr(B) - \Pr(AB) \tag{1.1.8.1}$$

$$\Pr(E) = \sum_{i=1}^{3} p_i q_i = \mathbf{p}^\top \mathbf{q} \tag{1.1.8.2}$$

$$\Pr(A_j|E) = \frac{p_j q_j}{\mathbf{p}^\top \mathbf{q}} \tag{1.1.8.3}$$

and A, B are independent if and only if $\Pr(AB) = \Pr(A)\Pr(B)$.

1.1.9 The conic $\mathbf{x}^\top \mathbf{V} \mathbf{x} = c$, with $\mathbf{V}$ symmetric positive definite having eigenvalues λ_1, λ_2, is an ellipse with semi-axes $\sqrt{c/\lambda_1}$ and $\sqrt{c/\lambda_2}$. Its area is

$$\frac{\pi c}{\sqrt{|\mathbf{V}|}} \tag{1.1.9.1}$$

1.1.10 A linear programming problem is

$$\max_{\mathbf{x}} \ \mathbf{c}^\top \mathbf{x} \quad \text{s.t.} \quad \mathbf{A}\mathbf{x} \preceq \mathbf{b}, \ \mathbf{x} \succeq \mathbf{0} \tag{1.1.10.1}$$

and, if the feasible region is a bounded polygon, the optimum is attained at a corner point. If two corner points give the same optimum, so does every point on the segment joining them.

1.1.11 Standard integrals,

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left|x + \sqrt{x^2+a^2}\right| + C \tag{1.1.11.1}$$

$$\int \sec x \, dx = \log|\sec x + \tan x| + C \tag{1.1.11.2}$$

1.1.12 A differentiable f is increasing on an interval if $f'(x) > 0$ there. For $y = u^v$,

$$\frac{dy}{dx} = y \frac{d}{dx}(v \log u) \tag{1.1.12.1}$$

1.1.13 For finite sets A, B with $|A| = m$, $|B| = n$, the number of relations from A to B is 2^{mn}. A bijection $A \to B$ exists only if $m = n$.

B. Problems

A (Questions 1–20, 1 mark each)

1.2.1 If the matrix $\mathbf{A} = \begin{pmatrix} -p & q \\ r & p \end{pmatrix}$ is such that $\mathbf{A}^2 = \mathbf{I}$, then

(A) $1 + p^2 + qr = 0$
(B) $1 - p^2 - qr = 0$
(C) $1 - p^2 + qr = 0$
(D) $1 + p^2 - qr = 0$

Solution:

$$\mathbf{A}^2 = \begin{pmatrix} -p & q \\ r & p \end{pmatrix}\begin{pmatrix} -p & q \\ r & p \end{pmatrix} = \begin{pmatrix} p^2+qr & 0 \\ 0 & p^2+qr \end{pmatrix} = \left(p^2+qr\right)\mathbf{I} \tag{1.2.1.1}$$

Hence $\mathbf{A}^2 = \mathbf{I}$ gives $p^2 + qr = 1$, i.e., $1 - p^2 - qr = 0$. Option (B). The following code verifies this and the solutions of Problems 2, 4 and 5.

codes/Matrix.py

1.2.2 If $\mathbf{A}$ is a square matrix such that $\mathbf{A}^2 = \mathbf{A}$, then $(\mathbf{A}-\mathbf{I})^3 - \mathbf{A}$ is equal to

(A) $\mathbf{I}$ (B) $-\mathbf{I}$ (C) $\mathbf{A}$ (D) $\mathbf{A}^2$

Solution: Since $\mathbf{A}^2 = \mathbf{A}$,

$$(\mathbf{A}-\mathbf{I})^2 = \mathbf{A}^2 - 2\mathbf{A} + \mathbf{I} = \mathbf{I} - \mathbf{A} \tag{1.2.2.1}$$

$$(\mathbf{A}-\mathbf{I})^3 = (\mathbf{A}-\mathbf{I})(\mathbf{I}-\mathbf{A}) = -(\mathbf{I}-\mathbf{A})^2 = \mathbf{A}-\mathbf{I} \tag{1.2.2.2}$$

$$\implies (\mathbf{A}-\mathbf{I})^3 - \mathbf{A} = -\mathbf{I} \tag{1.2.2.3}$$

Option (B).

1.2.3 For the inverse trigonometric functions, which of the following principal value branches is not correctly defined?

(A) $\tan^{-1} : \mathbb{R} \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(B) $\sec^{-1} : \mathbb{R} - (-1,1) \to [0,\pi] - \left\{\frac{\pi}{2}\right\}$
(C) $\cot^{-1} : \mathbb{R} \to (0,\pi)$
(D) $\text{cosec}^{-1} : \mathbb{R} - (-1,1) \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Solution: $\text{cosec}\, x = \frac{1}{\sin x}$ is undefined at $\sin x = 0$, so $x = 0$ must be removed from the range. The correct branch is

$$\text{cosec}^{-1} : \mathbb{R} - (-1,1) \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\} \tag{1.2.3.1}$$

The other three branches are the standard ones. Option (D).

1.2.4 Let $\mathbf{A} = \begin{pmatrix} 0 & -3 & 4 \\ 1 & 0 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} -3 & 0 & 1 \\ 2 & 4 & 0 \end{pmatrix}$. If $\mathbf{A}+\mathbf{B}+\mathbf{C} = \mathbf{O}$, then the matrix $\mathbf{C}$ is

(A) $\begin{pmatrix} -3 & -3 & 5 \\ 3 & 4 & 2 \end{pmatrix}$
(B) $\begin{pmatrix} 3 & 3 & 5 \\ -3 & -4 & -2 \end{pmatrix}$
(C) $\begin{pmatrix} 3 & 3 & -5 \\ -3 & -4 & -2 \end{pmatrix}$
(D) $\begin{pmatrix} -3 & -3 & -5 \\ 3 & 4 & 2 \end{pmatrix}$

Solution:

$$\mathbf{C} = -(\mathbf{A}+\mathbf{B}) = -\begin{pmatrix} -3 & -3 & 5 \\ 3 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 3 & -5 \\ -3 & -4 & -2 \end{pmatrix} \tag{1.2.4.1}$$

Option (C).

1.2.5 If $\mathbf{A}$ is non-singular, which of the following is not true?

(A) $\text{adj}\,\mathbf{A}$ is singular
(B) $(\text{adj}\,\mathbf{A})^{-1} = \text{adj}\left(\mathbf{A}^{-1}\right)$
(C) $\left|\mathbf{A}\right| \neq 0$
(D) $\mathbf{A}^{-1}$ exists

Solution: For a non-singular $\mathbf{A}$ of order n, from (1.1.1.2),

$$\left|\text{adj}\,\mathbf{A}\right| = \left|\mathbf{A}\right|^{n-1} \neq 0 \tag{1.2.5.1}$$

so $\text{adj}\,\mathbf{A}$ is non-singular and statement (A) is false. Option (A).

1.2.6 If $f(x) = \begin{cases} \dfrac{x^2-4x-5}{x+1}, & x \neq -1 \\ k, & x = -1 \end{cases}$ is continuous at $x = -1$, then the value of k is

(A) Any real value
(B) 6
(C) -1
(D) -6

Solution: Since $x^2 - 4x - 5 = (x-5)(x+1)$, continuity requires

$$k = \lim_{x\to -1} \frac{(x-5)(x+1)}{x+1} = \lim_{x\to -1} (x-5) = -6 \tag{1.2.6.1}$$

Option (D).

1.2.7 If the area of $\triangle ABC$ with vertices $\mathbf{A} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 0 \\ k \end{pmatrix}$ is 5 sq. units, then the values of k are

(A) $3, 1$
(B) $-1, 3$
(C) $-1, 2$
(D) $0, 2$

Solution:

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} -5 \\ 0 \end{pmatrix}, \quad \mathbf{C} - \mathbf{A} = \begin{pmatrix} -3 \\ k-1 \end{pmatrix} \tag{1.2.7.1}$$

From (1.1.3.1),

$$\frac{1}{2}\left|\begin{vmatrix} -5 & -3 \\ 0 & k-1 \end{vmatrix}\right| = \frac{5}{2}\left|k-1\right| = 5 \tag{1.2.7.2}$$

$$\implies k - 1 = \pm 2 \implies k = 3 \text{ or } -1 \tag{1.2.7.3}$$

Option (B). The code in Problem 25 verifies this.

1.2.8 The derivative of $\cos^{-1}\left(\dfrac{\sin x + \cos x}{\sqrt{2}}\right)$, $-\dfrac{\pi}{4} < x < \dfrac{\pi}{4}$, with respect to x is

(A) -1
(B) 1
(C) $\frac{\pi}{4}$
(D) $-\frac{\pi}{4}$

Solution:

$$\frac{\sin x + \cos x}{\sqrt{2}} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \sin x \\ \cos x \end{pmatrix} = \cos x \cos\frac{\pi}{4} + \sin x \sin\frac{\pi}{4} = \cos\left(x - \frac{\pi}{4}\right) \tag{1.2.8.1}$$

For $-\frac{\pi}{4} < x < \frac{\pi}{4}$, we have $x - \frac{\pi}{4} \in \left(-\frac{\pi}{2}, 0\right)$, so

$$\cos^{-1}\cos\left(x - \frac{\pi}{4}\right) = \frac{\pi}{4} - x \implies \frac{d}{dx}\left(\frac{\pi}{4} - x\right) = -1 \tag{1.2.8.2}$$

Option (A).

1.2.9 The absolute minimum value of $f(x) = (x-2)^2 + 5$ in the interval $[-3, 2]$ is

(A) -3 (B) 2 (C) 5 (D) 30

Solution:

$$f'(x) = 2(x-2) < 0 \quad \text{for } x \in [-3, 2) \tag{1.2.9.1}$$

So f is decreasing on $[-3, 2]$ and the minimum is at the right end, $f(2) = 5$. Option (C).

1.2.10 $\int \frac{1}{\sqrt{1+\cos 2x}}\,dx$ is equal to

(A) $\log\cos x + C$
(B) $\frac{1}{\sqrt{2}}\log|\sec x + \tan x| + C$
(C) $\frac{1}{\sqrt{2}}\log|\sec x - \tan x| + C$
(D) $\log\sin 2x + C$

Solution: Since $1 + \cos 2x = 2\cos^2 x$, taking $\cos x > 0$,

$$\int \frac{dx}{\sqrt{2}\cos x} = \frac{1}{\sqrt{2}}\int \sec x\,dx = \frac{1}{\sqrt{2}}\log|\sec x + \tan x| + C \tag{1.2.10.1}$$

from (1.1.11.2). Option (B).

1.2.11 The value of $\int_{-5}^{-1} \frac{1}{x}\,dx$ is equal to

(A) $-\log 5$ (B) x^6 (C) $\log(-5)$ (D) x^{-6}

Solution:

$$\int_{-5}^{-1} \frac{dx}{x} = [\log|x|]_{-5}^{-1} = \log 1 - \log 5 = -\log 5 \tag{1.2.11.1}$$

Option (A).

1.2.12 An ant crawls on a sheet of paper along the line $y = 2x - 4$. The area of the surface bounded by the y-axis, the x-axis and $x = 1$ is

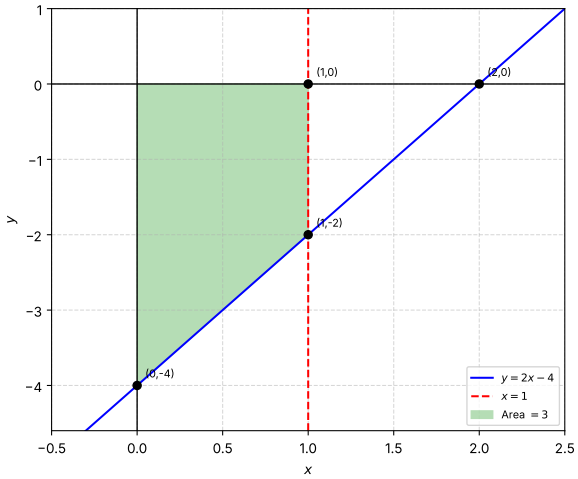
(A) 1 sq. unit (B) 3 sq. units (C) 2 sq. units (D) 4 sq. units

Solution: For $0 \le x \le 1$ the line lies below the x-axis (y goes from -4 to -2), so

$$\text{Area} = \int_0^1 |2x - 4|\,dx = \int_0^1 (4 - 2x)\,dx = \left[4x - x^2\right]_0^1 = 3 \tag{1.2.12.1}$$

This is a trapezium with parallel sides 4 and 2 and width 1, i.e., $\frac{4+2}{2}\cdot 1 = 3$. Option (B). The following code verifies and plots the graph

codes/Ant.py



1.2.13 The general solution of the differential equation $\frac{dy}{dx} = e^{3x-y}$ is

(A) $3e^y = e^{3x} + C$
(C) $e^{3x-y} = C$
(B) $\log(3x - y) = C$
(D) $-e^y + 3e^{3x} = C$

Solution: Separating the variables,

$$e^y\,dy = e^{3x}\,dx \implies e^y = \frac{e^{3x}}{3} + C' \implies 3e^y = e^{3x} + C \tag{1.2.13.1}$$

Option (A).

1.2.14 If the position vector $\mathbf{p}$ of the point $(24, n)$ satisfies $\|\mathbf{p}\| = 25$, then n is

(A) ± 49 (B) ± 5 (C) ± 1 (D) ± 7

Solution:

$$\|\mathbf{p}\|^2 = \begin{pmatrix}24 & n\end{pmatrix}\begin{pmatrix}24\\ n\end{pmatrix} = 576 + n^2 = 625 \implies n^2 = 49 \implies n = \pm 7 \tag{1.2.14.1}$$

Option (D).

1.2.15 The vectors $\mathbf{a} = \begin{pmatrix}3\\2\\\lambda\end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix}2\\-4\\5\end{pmatrix}$ represent the two strips of the Red Cross sign outside a doctor's clinic. The value of λ is

(A) 1 (B) $\frac{5}{2}$ (C) $\frac{2}{5}$ (D) 0

Solution: The strips of the cross are perpendicular. From (1.1.4.2),

$$\mathbf{a}^\top \mathbf{b} = 6 - 8 + 5\lambda = 0 \implies \lambda = \frac{2}{5} \tag{1.2.15.1}$$

Option (C).

1.2.16 If $3\Pr(A) = \Pr(B) = \frac{3}{5}$ and $\Pr(A|B) = \frac{2}{3}$, then $\Pr(A+B)$ is

(A) $\frac{3}{5}$ (B) $\frac{1}{5}$ (C) $\frac{2}{15}$ (D) $\frac{2}{5}$

Solution: Here $\Pr(A) = \frac{1}{5}$ and $\Pr(B) = \frac{3}{5}$. From (1.1.8.1),

$$\Pr(AB) = \Pr(A|B)\Pr(B) = \frac{2}{3}\cdot\frac{3}{5} = \frac{2}{5} \tag{1.2.16.1}$$

$$\Pr(A+B) = \frac{1}{5} + \frac{3}{5} - \frac{2}{5} = \frac{2}{5} \tag{1.2.16.2}$$

Option (D). Note that the given data are not consistent, since $\Pr(AB) > \Pr(A)$; the answer follows from the formula as the paper intends. The code in Problem 31 verifies this.

B (Questions 21–25, 2 marks each)

1.2.21 Evaluate $\sin\left[\tan^{-1}\left(\tan\frac{3\pi}{4}\right)\right]$.

Solution: $\tan\frac{3\pi}{4} = -1$ and the principal value of $\tan^{-1}$ lies in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, so

$$\tan^{-1}(-1) = -\frac{\pi}{4} \implies \sin\left(-\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} \tag{1.2.16.3}$$

1.2.22 (a) Differentiate x^x with respect to $x\log x$.

OR

(b) If $y = P\cos ux + Q\sin ux$, show that $\frac{d^2y}{dx^2} + u^2 y = 0$.

Solution: (a) Let $p = x^x$ and $q = x\log x$ for $x > 0$. Then $\log p = x\log x = q$, and from (1.1.12.1),

$$\frac{dp}{dx} = x^x(1+\log x), \quad \frac{dq}{dx} = 1 + \log x \tag{1.2.16.4}$$

$$\implies \frac{dp}{dq} = \frac{dp/dx}{dq/dx} = x^x \tag{1.2.16.5}$$

(b) Differentiating twice,

$$\frac{dy}{dx} = -Pu\sin ux + Qu\cos ux \tag{1.2.16.6}$$

$$\frac{d^2y}{dx^2} = -Pu^2\cos ux - Qu^2\sin ux = -u^2 y \tag{1.2.16.7}$$

$$\implies \frac{d^2y}{dx^2} + u^2 y = 0 \tag{1.2.16.8}$$

Remark: this is the equation of a sinusoidal signal of angular frequency u, e.g., the voltage across an ideal LC tank.

1.2.23 Determine the values of x for which $f(x) = \dfrac{x-3}{x+1}$, $x \neq -1$, is an increasing function.

Solution:

$$f'(x) = \frac{(x+1)-(x-3)}{(x+1)^2} = \frac{4}{(x+1)^2} > 0 \quad \forall x \neq -1 \tag{1.2.16.9}$$

Since $f'(x) > 0$, f is increasing on $(-\infty, -1)$ and on $(-1, \infty)$, i.e., for all $x \in \mathbb{R} - \{-1\}$.

1.2.24 (a) Three honey bees fly along the vectors $\mathbf{a} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 0 \\ 4 \\ -2 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}$. Find λ such that the path for $\mathbf{a} + \lambda\mathbf{b}$ is perpendicular to $\mathbf{c}$.

OR

(b) If $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are non-collinear points with $\mathbf{AB} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\mathbf{AC} = \begin{pmatrix} 2 \\ -3 \\ 0 \end{pmatrix}$, find the area of $\triangle ABC$.

Solution: (a) From (1.1.4.2),

$$(\mathbf{a} + \lambda\mathbf{b})^\top \mathbf{c} = \mathbf{a}^\top \mathbf{c} + \lambda \mathbf{b}^\top \mathbf{c} = 8 - 4\lambda = 0 \implies \lambda = 2 \tag{1.2.16.10}$$

(b)

$$\mathbf{AB} \times \mathbf{AC} = \begin{pmatrix} (2)(0) - (-1)(-3) \\ (-1)(2) - (1)(0) \\ (1)(-3) - (2)(2) \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \\ -7 \end{pmatrix} \tag{1.2.16.11}$$

From (1.1.3.1),

$$\text{Area} = \frac{1}{2}\sqrt{9+4+49} = \frac{\sqrt{62}}{2} \text{ sq. units} \tag{1.2.16.12}$$

1.2.25 Find the angle between the lines

$$\frac{x-2}{3} = \frac{y+5}{2} = \frac{1-z}{-6}, \qquad \frac{x-7}{1} = \frac{y}{2} = \frac{6-z}{-2} \tag{1.2.16.13}$$

Solution: Since $\frac{1-z}{-6} = \frac{z-1}{6}$ and $\frac{6-z}{-2} = \frac{z-6}{2}$, the direction vectors are

$$\mathbf{m}_1 = \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix}, \quad \mathbf{m}_2 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad \|\mathbf{m}_1\| = 7, \ \|\mathbf{m}_2\| = 3 \tag{1.2.16.14}$$

From (1.1.5.1),

$$\cos\theta = \frac{3+4+12}{7 \cdot 3} = \frac{19}{21} \implies \theta = \cos^{-1}\left(\frac{19}{21}\right) \approx 25.2^\circ \tag{1.2.16.15}$$

The following code verifies this and Problems 7, 16, 17, 24, 30 and 35.

codes/Lines.py

C (Questions 26–31, 3 marks each)

1.2.26 A spherical balloon loses its volume due to escape of air in such a way that the decrease of volume at any instant is proportional to its surface area. Show that the radius is decreasing at a constant rate.

Solution: Let $V = \frac{4}{3}\pi r^3$ and $S = 4\pi r^2$. For some constant $k > 0$,

$$\frac{dV}{dt} = -kS \implies 4\pi r^2 \frac{dr}{dt} = -k\left(4\pi r^2\right) \implies \frac{dr}{dt} = -k \tag{1.2.16.16}$$

which is a constant. Hence the radius decreases at a constant rate k.

1.2.27 (a) Find $\int \frac{x - \sin x}{1 - \cos x}\, dx$.

OR

(b) Evaluate $\int_0^2 \frac{1}{\sqrt{x^2 + 2x + 3}}\, dx$.

Solution: (a) Using $1 - \cos x = 2\sin^2 \frac{x}{2}$ and $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$,

$$\frac{x - \sin x}{1 - \cos x} = \frac{x}{2}\text{cosec}^2 \frac{x}{2} - \cot \frac{x}{2} \tag{1.2.16.17}$$

Integrating the first term by parts, with $u = x$ and $dv = \frac{1}{2}\text{cosec}^2 \frac{x}{2}\, dx$,

$$\int \frac{x}{2}\text{cosec}^2 \frac{x}{2}\, dx = -x \cot \frac{x}{2} + \int \cot \frac{x}{2}\, dx \tag{1.2.16.18}$$

The $\int \cot \frac{x}{2}\, dx$ terms cancel, so

$$\int \frac{x - \sin x}{1 - \cos x}\, dx = -x \cot \frac{x}{2} + C \tag{1.2.16.19}$$

(b) Since $x^2 + 2x + 3 = (x+1)^2 + 2$, from (1.1.11.1) with $a = \sqrt{2}$,

$$\int_0^2 \frac{dx}{\sqrt{(x+1)^2 + 2}} = \left[\log\left|(x+1) + \sqrt{x^2 + 2x + 3}\right|\right]_0^2 \tag{1.2.16.20}$$

$$= \log\left(3 + \sqrt{11}\right) - \log\left(1 + \sqrt{3}\right) = \log \frac{3 + \sqrt{11}}{1 + \sqrt{3}} \tag{1.2.16.21}$$

The following code verifies this and Problems 8, 10, 22, 23, 28 and 33.

codes/Calculus.py

1.2.28 Solve the differential equation $\left(x + 2y^3\right)\, dy = y\, dx$.

Solution: Treating x as a function of y,

$$\frac{dx}{dy} - \frac{1}{y}x = 2y^2 \tag{1.2.16.22}$$

This is linear with integrating factor $e^{-\int dy/y} = \frac{1}{y}$. Hence

$$\frac{d}{dy}\left(\frac{x}{y}\right) = 2y \implies \frac{x}{y} = y^2 + C \implies x = y^3 + Cy \tag{1.2.16.23}$$

1.2.29 (a) Three toys A, B and C are placed on a straight line. If their position vectors are $\begin{pmatrix}55\\-2\end{pmatrix}$, $\begin{pmatrix}5\\8\end{pmatrix}$ and $\begin{pmatrix}a\\-52\end{pmatrix}$ respectively, find a.

OR

(b) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are unit vectors, prove that $\|\mathbf{a}-\mathbf{b}\|^2 + \|\mathbf{b}-\mathbf{c}\|^2 + \|\mathbf{c}-\mathbf{a}\|^2 \leq 9$.

Solution: (a) From (1.1.3.2),

$$\left|\mathbf{B}-\mathbf{A} \quad \mathbf{C}-\mathbf{A}\right| = \begin{vmatrix}-50 & a-55\\ 10 & -50\end{vmatrix} = 2500 - 10\,(a-55) = 0 \implies a = 305 \tag{1.2.16.24}$$

(b) Let $s = \mathbf{a}^\top\mathbf{b} + \mathbf{b}^\top\mathbf{c} + \mathbf{c}^\top\mathbf{a}$. Since the vectors are unit vectors, from (1.1.4.1),

$$\|\mathbf{a}-\mathbf{b}\|^2 + \|\mathbf{b}-\mathbf{c}\|^2 + \|\mathbf{c}-\mathbf{a}\|^2 = 6 - 2s \tag{1.2.16.25}$$

But

$$0 \leq \|\mathbf{a}+\mathbf{b}+\mathbf{c}\|^2 = 3 + 2s \implies s \geq -\frac{3}{2} \tag{1.2.16.26}$$

Therefore

$$6 - 2s \leq 6 + 3 = 9 \tag{1.2.16.27}$$

1.2.30 (a) A die is rolled. For the events $A = \{1,2,5\}$, $B = \{3,5\}$, $C = \{2,3,4,5\}$ find (i) $\Pr(A|C)$ and $\Pr(C|A)$, (ii) $\Pr(AB|C)$ and $\Pr(A+B|C)$.

OR

(b) A box contains cards numbered 1 to 6. Two cards are picked one by one with replacement. Let A be the event that the sum of the numbers is 10 and B the event that the first card is not 4. Find $\Pr(AB)$ and state whether A and B are independent.

Solution: (a) Each outcome has probability $\frac{1}{6}$ and $AC = \{2,5\}$, $AB = \{5\}$, $(A+B)\,C = \{2,3,5\}$. From (1.1.8.1),

$$\Pr(A|C) = \frac{2/6}{4/6} = \frac{1}{2}, \quad \Pr(C|A) = \frac{2/6}{3/6} = \frac{2}{3} \tag{1.2.16.28}$$

$$\Pr(AB|C) = \frac{1/6}{4/6} = \frac{1}{4}, \quad \Pr(A+B|C) = \frac{3/6}{4/6} = \frac{3}{4} \tag{1.2.16.29}$$

(b) There are 36 equally likely outcomes. $A = \{(4,6),(5,5),(6,4)\}$ and $AB = \{(5,5),(6,4)\}$, while B has $5 \times 6 = 30$ outcomes. So

$$\Pr(A) = \frac{1}{12}, \quad \Pr(B) = \frac{5}{6}, \quad \Pr(AB) = \frac{2}{36} = \frac{1}{18} \tag{1.2.16.30}$$

Now $\Pr(A)\Pr(B) = \frac{5}{72} \neq \frac{4}{72} = \Pr(AB)$, so A and B are not independent. The following code verifies this and Problems 18 and 37.

codes/Prob.py

D (Questions 32–35, 5 marks each)

1.2.32 A man buys fruits. 4 apples, 3 oranges and 2 bananas cost ₹ 60; 2 apples, 4 oranges and 6 bananas cost ₹ 90; 6 apples, 2 oranges and 3 bananas cost ₹ 70. Using the matrix method, find the cost of one fruit of each kind.

Solution: Let the prices of an apple, an orange and a banana be x_1, x_2, x_3. Then $\mathbf{Ax} = \mathbf{y}$ with

$$\mathbf{A} = \begin{pmatrix} 4 & 3 & 2 \\ 2 & 4 & 6 \\ 6 & 2 & 3 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 60 \\ 90 \\ 70 \end{pmatrix} \tag{1.2.16.31}$$

Expanding along the first row,

$$\left|\mathbf{A}\right| = 4\left(12-12\right) - 3\left(6-36\right) + 2\left(4-24\right) = 0 + 90 - 40 = 50 \neq 0 \tag{1.2.16.32}$$

The cofactors are $C_{11} = 0$, $C_{12} = 30$, $C_{13} = -20$, $C_{21} = -5$, $C_{22} = 0$, $C_{23} = 10$, $C_{31} = 10$, $C_{32} = -20$, $C_{33} = 10$, so

$$\text{adj}\,\mathbf{A} = \begin{pmatrix} 0 & -5 & 10 \\ 30 & 0 & -20 \\ -20 & 10 & 10 \end{pmatrix} \tag{1.2.16.33}$$

From (1.1.1.1) and (1.1.2.1),

$$\mathbf{x} = \frac{1}{50}\begin{pmatrix} 0 & -5 & 10 \\ 30 & 0 & -20 \\ -20 & 10 & 10 \end{pmatrix}\begin{pmatrix} 60 \\ 90 \\ 70 \end{pmatrix} = \frac{1}{50}\begin{pmatrix} 250 \\ 400 \\ 400 \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ 8 \end{pmatrix} \tag{1.2.16.34}$$

An apple costs ₹ 5, an orange ₹ 8 and a banana ₹ 8. The following code verifies this.

codes/Fruits.py

1.2.33 (a) If $y\sqrt{x^2+1} = \log\left(\sqrt{x^2+1} - x\right)$, show that $\left(x^2+1\right)\frac{dy}{dx} + xy + 1 = 0$.

OR

(b) Find the derivative of $x^{\cot x} + \dfrac{2x^2-3}{2x^2-x+2}$ with respect to x.

Solution: (a) Let $s = \sqrt{x^2+1}$, so $\frac{ds}{dx} = \frac{x}{s}$. Differentiating $ys = \log\left(s-x\right)$,

$$s\frac{dy}{dx} + \frac{xy}{s} = \frac{1}{s-x}\left(\frac{x}{s} - 1\right) = \frac{1}{s-x}\cdot\frac{x-s}{s} = -\frac{1}{s} \tag{1.2.16.35}$$

Multiplying by s and using $s^2 = x^2+1$,

$$\left(x^2+1\right)\frac{dy}{dx} + xy + 1 = 0 \tag{1.2.16.36}$$

(b) Let $u = x^{\cot x}$ and $v = \frac{2x^2-3}{2x^2-x+2}$. From (1.1.12.1),

$$\frac{du}{dx} = x^{\cot x}\frac{d}{dx}\left(\cot x \log x\right) = x^{\cot x}\left(\frac{\cot x}{x} - \text{cosec}^2 x \log x\right) \tag{1.2.16.37}$$

$$\frac{dv}{dx} = \frac{4x\left(2x^2-x+2\right) - \left(2x^2-3\right)(4x-1)}{\left(2x^2-x+2\right)^2} = \frac{-2x^2+20x-3}{\left(2x^2-x+2\right)^2} \tag{1.2.16.38}$$

Hence

$$\frac{d}{dx}(u+v) = x^{\cot x}\left(\frac{\cot x}{x} - \text{cosec}^2 x \log x\right) + \frac{-2x^2+20x-3}{\left(2x^2-x+2\right)^2} \tag{1.2.16.39}$$

1.2.34 Sketch the curve $\left\{(x,y) : 100x^2 + 25y^2 = 2500\right\}$ and find the area of the region enclosed by it using integration.

Solution: The curve is $\mathbf{x}^\top \mathbf{V}\mathbf{x} = c$ with

$$\mathbf{V} = \begin{pmatrix} 100 & 0 \\ 0 & 25 \end{pmatrix}, \quad c = 2500 \tag{1.2.16.40}$$

The eigenvalues of $\mathbf{V}$ are 100 and 25, so from (1.1.9.1) it is the ellipse

$$\frac{x^2}{25} + \frac{y^2}{100} = 1 \tag{1.2.16.41}$$

with semi-axes 5 along the x-axis and 10 along the y-axis, symmetric about both axes (Fig. 1.2.16.1). In the first quadrant $y = 2\sqrt{25-x^2}$, so by symmetry

$$\text{Area} = 4\int_0^5 2\sqrt{25-x^2}\,dx = 8\left[\frac{x}{2}\sqrt{25-x^2} + \frac{25}{2}\sin^{-1}\frac{x}{5}\right]_0^5 \tag{1.2.16.42}$$

$$= 8 \cdot \frac{25}{2} \cdot \frac{\pi}{2} = 50\pi \text{ sq. units} \tag{1.2.16.43}$$

which agrees with $\frac{\pi c}{\sqrt{|\mathbf{V}|}} = \frac{2500\pi}{50} = 50\pi$ from (1.1.9.1). The following code computes the area and generates Fig. 1.2.16.1.

codes/Ellipse.py

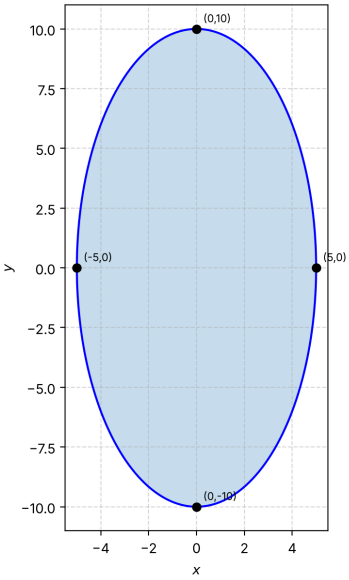


Fig. 1.2.16.1: $100x^2 + 25y^2 = 2500$

1.2.35 (a) Find the foot of the perpendicular from the point $\mathbf{Q} = \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$ on the line $\frac{-x-3}{-5} = \frac{1-y}{-2} = \frac{3z+12}{9}$ and hence the length of the perpendicular.

OR

(b) Find p if the shortest distance between the lines $\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ and $\mathbf{x} = \begin{pmatrix} p \\ -1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$ is $\frac{3}{\sqrt{2}}$ units.

Solution: (a) The line is $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$, i.e.,

$$\mathbf{x} = \mathbf{P}_0 + \lambda \mathbf{m}, \quad \mathbf{P}_0 = \begin{pmatrix} -3 \\ 1 \\ -4 \end{pmatrix}, \ \mathbf{m} = \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix} \tag{1.2.16.44}$$

From (1.1.6.1),

$$\lambda_0 = \frac{\begin{pmatrix} 5 & 2 & 3 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 7 \end{pmatrix}}{\begin{pmatrix} 5 & 2 & 3 \end{pmatrix} \begin{pmatrix} 5 \\ 2 \\ 3 \end{pmatrix}} = \frac{38}{38} = 1 \implies \mathbf{F} = \mathbf{P}_0 + \mathbf{m} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} \tag{1.2.16.45}$$

The length of the perpendicular is

$$\|\mathbf{F}-\mathbf{Q}\| = \left\| \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} \right\| = \sqrt{4+1+16} = \sqrt{21} \tag{1.2.16.46}$$

(b)

$$\mathbf{m}_1 \times \mathbf{m}_2 = \begin{pmatrix} (-1)(2)-(1)(1) \\ (1)(2)-(1)(2) \\ (1)(1)-(-1)(2) \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 3 \end{pmatrix}, \quad \|\mathbf{m}_1 \times \mathbf{m}_2\| = 3\sqrt{2} \tag{1.2.16.47}$$

With $\mathbf{A}_2 - \mathbf{A}_1 = \begin{pmatrix} p-1 \\ -3 \\ -2 \end{pmatrix}$, from (1.1.7.1),

$$d = \frac{|-3(p-1)-6|}{3\sqrt{2}} = \frac{|p+1|}{\sqrt{2}} = \frac{3}{\sqrt{2}} \implies p+1 = \pm 3 \implies p = 2 \text{ or } -4 \tag{1.2.16.48}$$

E (Questions 36–38, 4 marks each)

1.2.36 Case Study 2. Three vaccines A_1, A_2, A_3 were given to 25%, 35% and 40% of the population respectively. They protect against the infection with probabilities 60%, 55% and 50% respectively. Find the probability that

(i) a person taking vaccine A_2 gets infected. [1]

(ii) a randomly chosen person is protected from the infection. [1]

(iii) (a) the person was given vaccine A_1, given that the randomly chosen person is infected. [2]

OR

(b) the person was given vaccine A_3, given that the randomly chosen person is not infected. [2]

Solution: Let E be the event that the person is protected, and

$$\mathbf{p} = \begin{pmatrix} 0.25 \\ 0.35 \\ 0.40 \end{pmatrix}, \quad \mathbf{q} = \begin{pmatrix} 0.60 \\ 0.55 \\ 0.50 \end{pmatrix} \tag{1.2.16.49}$$

with $p_i = \Pr(A_i)$ and $q_i = \Pr(E|A_i)$.

(i) $\Pr(E'|A_2) = 1 - q_2 = 0.45 = \frac{9}{20}$.

(ii) From (1.1.8.2),

$$\Pr(E) = \mathbf{p}^\top \mathbf{q} = 0.15 + 0.1925 + 0.20 = 0.5425 = \frac{217}{400} \tag{1.2.16.50}$$

(iii) Here $\Pr(E') = 1 - \frac{217}{400} = \frac{183}{400}$. From (1.1.8.3),

(a)

$$\Pr\left(A_1|E'\right) = \frac{p_1\left(1-q_1\right)}{\Pr\left(E'\right)} = \frac{0.25\times 0.40}{0.4575} = \frac{40}{183} \approx 0.219 \tag{1.2.16.51}$$

(b)

$$\Pr\left(A_3|E\right) = \frac{p_3 q_3}{\mathbf{p}^\top \mathbf{q}} = \frac{0.40\times 0.50}{0.5425} = \frac{80}{217} \approx 0.369 \tag{1.2.16.52}$$

1.2.37 Case Study 3. A company produces cylindrical tumblers, open at the top, of fixed surface area. For a tumbler of volume V, height h and base radius r (Fig. 1.2.16.2),

(i) differentiate V with respect to r when the surface area is constant. [2]

(ii) if the volume is to be maximised, establish a relation between h and r. [2]

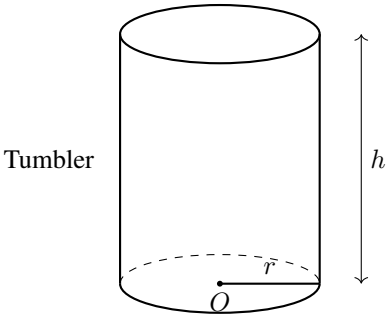


Fig. 1.2.16.2: Open cylindrical tumbler

Solution:

(i) The surface area (curved surface and base) is a constant S:

$$S = \pi r^2 + 2\pi r h \implies h = \frac{S-\pi r^2}{2\pi r} \implies V = \pi r^2 h = \frac{Sr-\pi r^3}{2} \tag{1.2.16.53}$$

$$\frac{dV}{dr} = \frac{S-3\pi r^2}{2} = \pi r\left(h-r\right) \tag{1.2.16.54}$$

where the last step substitutes $S = \pi r^2 + 2\pi r h$.

(ii) Setting $\frac{dV}{dr} = 0$ gives $h = r$ (equivalently $S = 3\pi r^2$). Also

$$\frac{d^2V}{dr^2} = -3\pi r < 0 \tag{1.2.16.55}$$

so V is maximum when $\boxed{h = r}$. Fig. 1.2.16.3 shows V against r for $S = 3\pi$, with the maximum at $r = h = 1$. The following code verifies this and generates Fig. 1.2.16.3.

codes/Tumbler.py

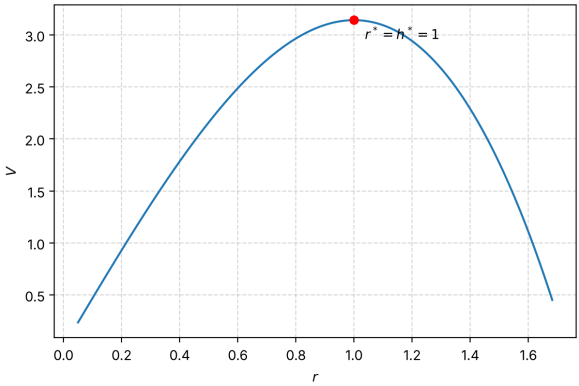


Fig. 1.2.16.3: V vs r for $S = 3\pi$